

Ismael Diaz

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EDUCATION

Duke University

B.S.E in Electrical and Computer Engineering and Computer Science

Durham, NC

Aug. 2023 – May 2027

EXPERIENCE

Undergraduate Teaching Assistant

ECE 250 - Computer Architecture

Duke University

Aug. 2024 – Present

- Led weekly recitations for 18 students; held office hours and developed exam review materials

ECE 350 - Digital Systems Lab

Jan. 2025 – Present

- Led weekly 12-student lab section; graded exams and mentored final projects

Cyber-Physical Systems Research Assistant

Duke University

Durham, NC

Aug. 2025 – Present

- Collecting AR hand-sensing dataset for open-source release (targetting 1M samples)
- Developing embedded software for mmWave radar integration on UAVs (Kria KR260 FPGA), enabling real-time sensing capabilities

Duke Electric Vehicles (DEV)

Member

Durham, NC

Aug. 2025 – Present

- Designed 12V→5V two-layer regulator PCB in Altium with noise filtering, used across all vehicle boards

Iglesia Bautista Libertad

Technology Assistant

Houston, TX

May 2023 – Present

- Deployed UniFi network with VLAN segmentation, improving reliability and security for 150+ devices
- Coordinated development of interactive missions display boards, increasing member engagement with missionaries
- Automated Planning Center reporting, eliminating manual workflows and saving around 4 hours/week

Amazon

Software Development Engineering Intern - Hub Delivery Team

Bellevue, WA

May 2025 – Aug. 2025

- Extended backend Java APIs to support new features for internal search tools used by thousands of store operators
- Designed chaos-testing strategy for a tier-1 service, raising Prime Day readiness score above 70%
- Increased unit test coverage and authored technical documentation to improve team onboarding

Software Development Engineering Intern - Connected Vehicle Team

May 2024 – Aug. 2024

- Developed 4 React dashboards for Amazon's Fleet Edge IoT platform, enabling VIN-based monitoring of vehicle location, device health, and telemetry
- Integrated telemetry APIs backed by AWS infrastructure to support real-time vehicle insights

PROJECTS

RISC-V Floating-Point Single-Precision Addition Approximation ISA Extension

Oct. 2025 – Dec. 2025

- Designed a custom Verilog IEEE-754-style approximate FP adder
- Evaluated accuracy-performance tradeoffs via gem5 simulations and microbenchmarks

FPGA Vending Machine

Jan. 2025 – May 2025

- Designed a 5-stage pipelined MIPS-like CPU with bypassing, mult/div, and memory-mapped I/O
- Deployed on Nexys A7 FPGA using Verilog, Vivado, and executing custom assembly programs

TECHNICAL SKILLS

Certifications: Oracle Java SE 8 Certified Associate, Autodesk Fusion 360

Languages: Java, Verilog, C, Python, MIPS Assembly, TypeScript, HTML/CSS, R

Frameworks: ROS2, gem5, React, Node.js, JUnit, WordPress, Joomla!

Developer Tools: Git, Docker, AWS, Google Cloud, Vivado, Vitis, CI/CD, IntelliJ, RStudio, Altium